Homework 2.1: Casino Probability, Sigma-Algebras, and Bad Decisions

Introduction

This homework uses roulette and blackjack to study probability spaces, sigma-algebras, events, random variables, expectation, variance, and decision-making under uncertainty.

The main goal is not to determine how to gamble successfully. Instead, the casino provides a simple setting in which mathematical probability models can be constructed and checked.

The analysis has four main components:

1. construct a probability space for European roulette;
2. construct a probability space for a simplified blackjack model;
3. use expected value to decide how a fixed bankroll should be allocated;
4. formulate and verify two additional probability questions.

An LLM was used to help formulate and analyze the models, but its predictions were checked using mathematical derivations and Monte Carlo simulations.

Part I: European Roulette

Probability Space

For European roulette, the sample space is

$$\Omega_R = \{0, 1, 2, \ldots, 36\}.$$

There are 37 possible outcomes.

Each outcome is equally likely, so

$$\mathbb{P}(\{\omega\}) = \frac{1}{37}, \qquad \omega \in \Omega_R.$$

Since the sample space is finite, the natural sigma-algebra is the full power set

$$\mathcal{F}_R = 2^{\Omega_R}.$$

Therefore, every subset of roulette outcomes is an event.

Explicitly Defined Roulette Events

The event that the roulette result is red is

$$A_{\text{red}} = \{1, 3, 5, 7, 9, 12, 14, 16, 18, 19, 21, 23, 25, 27, 30, 32, 34, 36\}.$$

There are 18 red numbers, so

$$\mathbb{P}(A_{\text{red}}) = \frac{18}{37}.$$

The event that the roulette result is even is

$$A_{\text{even}} = \{2, 4, 6, \ldots, 36\}.$$

There are also 18 even outcomes, so

$$\mathbb{P}(A_{\text{even}}) = \frac{18}{37}.$$

The event that the ball lands on zero is

$$A_{\text{zero}} = \{0\}.$$

Therefore,

$$\mathbb{P}(A_{\text{zero}}) = \frac{1}{37}.$$

These provide three explicitly defined measurable events in the roulette probability space.

Roulette Payoff Random Variable

For a one-unit bet on red, define

$$X_R(\omega) = \begin{cases} 1, & \omega \in A_{\text{red}}, \\ -1, & \omega \notin A_{\text{red}}. \end{cases}$$

Thus,

$$\mathbb{P}(X_R = 1) = \frac{18}{37}$$

and

$$\mathbb{P}(X_R = -1) = \frac{19}{37}.$$

The 19 losing outcomes are the 18 black numbers together with zero.

Expected Value

The expected payoff is

$$E[X_R] = 1\left(\frac{18}{37}\right) + (-1)\left(\frac{19}{37}\right).$$

Therefore,

$$E[X_R] = -\frac{1}{37}.$$

Numerically,

$$E[X_R] \approx -0.027027.$$

Thus a one-dollar red bet loses approximately 2.70 cents on average.

Variance

Since

$$X_R^2 = 1$$

for every outcome,

$$E[X_R^2] = 1.$$

Therefore,

$$\mathrm{Var}(X_R) = E[X_R^2] - (E[X_R])^2.$$

Hence,

$$\mathrm{Var}(X_R) = 1 - \left(-\frac{1}{37}\right)^2 = \frac{1368}{1369}.$$

Numerically,

$$\mathrm{Var}(X_R) \approx 0.999270.$$

Monte Carlo Verification

I independently simulated

$$1000000$$

European roulette spins.

The theoretical expected payoff was

$$-0.0270270,$$

while the simulated expected payoff was

$$-0.028258.$$

The difference was

$$-0.001231.$$

The Monte Carlo standard error was approximately

$$0.0009996.$$

Therefore, the difference was approximately

$$-1.23$$

standard errors from the theoretical value.

The theoretical variance was

$$0.999270,$$

while the simulated variance was

$$0.999201.$$

The simulation therefore agrees well with the analytical probability model.

Roulette Bankroll Decision

Suppose the player has bankroll B and chooses to bet b_R dollars on roulette.

Then

$$0 \leq b_R \leq B.$$

The total roulette profit is

$$\Pi_R = b_R X_R.$$

Therefore,

$$E[\Pi_R] = b_R E[X_R].$$

Since

$$E[X_R] = -\frac{1}{37},$$

we obtain

$$E[\Pi_R] = -\frac{b_R}{37}.$$

This decreases as b_R increases.

Therefore, the expected-value maximizing decision is

$$b_R^* = 0.$$

For a bankroll of

$$B = 100,$$

the theoretical and simulated results were:

Roulette Bet	Theoretical Expected Profit	Simulated Mean Profit
0	0.0000	0.0000
25	-0.6757	-0.7065
50	-1.3514	-1.4129
75	-2.0270	-2.1194
100	-2.7027	-2.8258



Both theory and simulation show that increasing the roulette bet decreases expected profit.

Therefore, the optimal roulette decision is to bet nothing.

Part II: Simplified Blackjack

Probability Space

For blackjack, the sample space is

$$\Omega_B = \{\text{all permutations of a standard 52-card deck}\}.$$

The cards can be regarded as 52 distinct physical cards, so

$$|\Omega_B| = 52!.$$

Assuming a perfectly shuffled deck,

$$\mathbb{P}(\{\omega\}) = \frac{1}{52!}.$$

The natural sigma-algebra is

$$\mathcal{F}_B = 2^{\Omega_B}.$$

Once the deck ordering and player strategy are fixed, the complete hand is determined.

Simplified Blackjack Rules

The model used in this report has the following rules:

- one standard 52-card deck;
- the initial deal is player card, dealer card, player card, dealer card;
- no splitting;
- no doubling;
- no surrender;
- player hits below 17;
- player stands on 17 or greater;
- dealer hits below 17;
- dealer stands on 17 or greater;
- an ace counts as 11 unless this would cause a bust, in which case it counts as 1;
- win gives payoff $+1$;
- loss gives payoff -1;
- push gives payoff 0.

The model treats all totals of 17 as standing totals, including soft 17.

The homework does not specify a special payout for a natural blackjack. Therefore, a natural blackjack is treated as an ordinary total of 21 rather than receiving a special $3:2$ payoff.

This is a modeling assumption and must be stated explicitly.

The sample points in Ω_B are permutations of 52 distinct physical cards. In the probability

calculation, cards with the same blackjack value can be grouped because suit does not affect the strategy or payoff. The multiplicities of the remaining card values are nevertheless updated after every draw, so the calculation still represents drawing without replacement.

Fixed Player Strategy

The fixed strategy is

$$S = \begin{cases} \text{hit}, & \text{total} < 17, \\ \text{stand}, & \text{total} \geq 17. \end{cases}$$

This strategy was chosen for simplicity and is not claimed to be optimal blackjack strategy.

Blackjack Events

Define the blackjack win event

$$A_{\text{win}} = \{\omega \in \Omega_B : \text{the player wins under strategy } S\}.$$

Similarly, define

$$A_{\text{loss}} = \{\omega \in \Omega_B : \text{the player loses}\}$$

and

$$A_{\text{push}} = \{\omega \in \Omega_B : \text{the final result is a tie}\}.$$

The blackjack probabilities were computed recursively while keeping track of the remaining deck. If a particular card value has n_j copies remaining out of N total cards, then

$$\mathbb{P}(\text{next card has value } j \mid \text{current state}) = \frac{n_j}{N}.$$

After each draw, the appropriate count is reduced by one. The recursion continues until the player and dealer rules determine a win, loss, or push. This is an exact finite-deck calculation: it accounts for dependence between successive draws and is equivalent to summing the payoff over all ordered decks, while exploiting the fact that cards with the same blackjack value are interchangeable for this simplified strategy.

The exact recursive calculation gave

$$\mathbb{P}(A_{\text{win}}) = 0.4088236578,$$

$$\mathbb{P}(A_{\text{loss}}) = 0.4889154379,$$

and

$$\mathbb{P}(A_{\text{push}}) = 0.1022609043.$$

These probabilities satisfy

$$\mathbb{P}(A_{\text{win}}) + \mathbb{P}(A_{\text{loss}}) + \mathbb{P}(A_{\text{push}}) = 1.$$

Blackjack Payoff Random Variable

Define

$$X_B : \Omega_B \to \mathbb{R}$$

by

$$X_B(\omega) = \begin{cases} 1, & \omega \in A_{\text{win}}, \\ 0, & \omega \in A_{\text{push}}, \\ -1, & \omega \in A_{\text{loss}}. \end{cases}$$

This is the second main payoff random variable used in the report.

Expected Value

The expected payoff is

$$E[X_B] = \mathbb{P}(A_{\text{win}}) - \mathbb{P}(A_{\text{loss}}).$$

Therefore,

$$E[X_B] = 0.4088236578 - 0.4889154379.$$

Hence,

$$E[X_B] = -0.0800917801.$$

Thus this simplified blackjack strategy loses approximately

$$8.01\%$$

of the amount bet on average.

Variance

Since

$$X_B^2 = \begin{cases} 1, & \text{win or loss}, \\ 0, & \text{push}, \end{cases}$$

we have

$$E[X_B^2] = \mathbb{P}(A_{\text{win}}) + \mathbb{P}(A_{\text{loss}}).$$

Thus,

$$E[X_B^2] = 0.8977390957.$$

Therefore,

$$\text{Var}(X_B) = E[X_B^2] - (E[X_B])^2.$$

Numerically,

$$\text{Var}(X_B) \approx 0.8913244025.$$

Blackjack Monte Carlo Verification

I independently simulated

$$500000$$

blackjack hands.

The theoretical probabilities were

$$\mathbb{P}(\text{win}) = 0.408824,$$

$$\mathbb{P}(\text{loss}) = 0.488915,$$

and

$$\mathbb{P}(\text{push}) = 0.102261.$$

The simulation produced

$$P_{\text{sim}}(\text{win}) = 0.409032,$$

$$P_{\text{sim}}(\text{loss}) = 0.488478,$$

and

$$P_{\text{sim}}(\text{push}) = 0.102490.$$

The theoretical expected payoff was

$$-0.0800918,$$

while the simulated expected payoff was

$$-0.079446.$$

The difference was only approximately

$$0.48$$

Monte Carlo standard errors.

The theoretical variance was

$$0.8913244,$$

while the simulated variance was

$$0.8911983.$$

Thus the simulation closely agrees with the recursive probability calculation.

Blackjack Bankroll Decision

Suppose the player bets b_B dollars on blackjack.

Then

$$0 \le b_B \le B.$$

The blackjack profit is

$$\Pi_B = b_B X_B.$$

Therefore,

$$E[\Pi_B] = b_B E[X_B].$$

Since

$$E[X_B] \approx -0.0800918,$$

we obtain

$$E[\Pi_B] \approx -0.0800918 b_B.$$

This is a decreasing function of the amount bet.

Therefore,

$$b_B^* = 0.$$

For

$$B = 100,$$

the numerical results were:

Blackjack Bet	Theoretical Expected Profit	Simulated Mean Profit
0	0.0000	0.0000
25	-2.0023	-1.9862
50	-4.0046	-3.9723
75	-6.0069	-5.9585
100	-8.0092	-7.9446



Therefore, under this simplified strategy, the expected-value maximizing decision is also to bet nothing.

Part III: Expected Value and Bankroll Allocation

The player now chooses both

$$b_R \geq 0$$

and

$$b_B \geq 0$$

subject to

$$b_R + b_B \leq B.$$

The expected total profit is

$$E[\Pi] = b_R e_R + b_B e_B.$$

The optimization problem is

$$\max_{b_R, b_B \geq 0,\, b_R + b_B \leq B} \left(b_R e_R + b_B e_B \right).$$

The feasible set is the triangle with vertices $(0,0)$, $(B,0)$, and $(0,B)$. Because the objective is linear, a maximum is attained at a vertex (or along an edge in a tie case). This makes the required sign cases transparent.

Case 1: $e_R < 0$ and $e_B < 0$

Both games lose money on average.

Any positive b_R or b_B reduces expected profit.

Since the player may choose not to bet,

$$b_R^* = 0, \qquad b_B^* = 0.$$

**## Case 2: $e_R < e_B * *_< 0$_

Blackjack is better than roulette in this hypothetical case, but both expected values remain negative.

Because not betting has expected return zero,

$$0 > e_B > e_R.$$

Therefore the optimum is still

$$b_R^* = 0, \qquad b_B^* = 0.$$

Being less bad than another game does not make a game a good expected-value decision.

Case 3: $e_R < 0 < e_B$

Roulette has negative expectation while blackjack has positive expectation.

Therefore no money should be placed into roulette:

$$b_R^* = 0.$$

Since each dollar placed into blackjack increases expected profit,

$$b_B^* = B.$$

Thus,

$$b_R^* = 0, \qquad b_B^* = B.$$

Application to My Results

For roulette,

$$e_R = -\frac{1}{37} \approx -0.0270270.$$

For my simplified blackjack strategy,

$$e_B \approx -0.0800918.$$

Thus,

$$e_B < e_R < 0.$$

Although roulette is better than this simplified blackjack strategy,

$$0 > e_R > e_B.$$

Therefore the optimal decision is

$$b_R^* = 0, \qquad b_B^* = 0.$$

For

$$B = 100,$$

the maximum expected profit is

$$E[\Pi]^* = 0.$$

Independent Bankroll Verification

A standalone simulation reconstructed both games from scratch, compared representative feasible allocations, and also searched a grid of bankroll allocations.

The theoretical and simulated strategy comparison was:

Strategy	Theoretical Profit	Simulated Profit
Bet nothing	0.0000	0.0000
All roulette	-2.7027	-3.2413
75 roulette / 25 blackjack	-4.0293	-4.3117
Half roulette / half blackjack	-5.3559	-5.3820
25 roulette / 75 blackjack	-6.6826	-6.4523
All blackjack	-8.0092	-7.5227



The theoretical grid search selected

$$b_R = 0, \qquad b_B = 0.$$

The independent Monte Carlo grid search also selected

$$b_R = 0, \qquad b_B = 0.$$

Therefore the mathematical optimization and numerical simulation agree.

Part IV: Two Additional Probability Questions

Question 1: Probability of Losing Five Roulette Bets in a Row

Suppose a player bets on red for five independent roulette spins.

For five spins, the sample space is

$$\Omega_5 = \Omega_R^5.$$

Because the spins are independent and each single-spin outcome is uniform, each ordered five-spin sequence has probability

$$\left(\frac{1}{37}\right)^5.$$

The natural sigma-algebra is the full power set 2^{Ω_5}.

Define

$$L_5 = \{\text{all five red bets lose}\}.$$

Since a red bet loses with probability

$$\frac{19}{37},$$

independence gives

$$\mathbb{P}(L_5) = \left(\frac{19}{37}\right)^5.$$

Numerically,

$$\mathbb{P}(L_5) = 0.0357074950.$$

Thus,

$$\mathbb{P}(L_5) \approx 3.57075\%.$$

A Monte Carlo simulation of

$$500000$$

five-spin experiments gave

$$P_{\text{sim}}(L_5) = 0.035462.$$

The difference from theory was approximately

$$-0.94$$

Monte Carlo standard errors.

Therefore, the simulation agrees with the theoretical result.

Question 2: Expected Profit After 100 Roulette Bets

Consider 100 independent one-unit bets on red. The sample space is

$$\Omega_{100} = \Omega_R^{100}.$$

Because the spins are independent and uniform, each ordered sequence in Ω_{100} has probability 37^{-100}. Since the space is finite, the natural sigma-algebra is

$$\mathcal{F}_{100} = 2^{\Omega_{100}}.$$

Let X_i be the payoff from the i-th one-unit red bet, with $X_i = 1$ for a win and $X_i = -1$ for a loss. Define the total profit

$$S_{100} = X_1 + \cdots + X_{100}.$$

For every i,

$$E[X_i] = -\frac{1}{37}.$$

By linearity of expectation,

$$E[S_{100}] = \sum_{i=1}^{100} E[X_i] = 100\left(-\frac{1}{37}\right) = -\frac{100}{37} \approx -2.70270.$$

For the variance, independence matters. Since the 100 roulette spins are independent,

$$\mathrm{Var}(S_{100}) = \sum_{i=1}^{100} \mathrm{Var}(X_i) = 100\left(\frac{1368}{1369}\right) \approx 99.92695.$$

The same result can be verified with a binomial model. Let W be the number of winning red bets. Then

$$W \sim \mathrm{Binomial}\left(100, \frac{18}{37}\right),$$

and

$$S_{100} = W - (100 - W) = 2W - 100.$$

Hence,

$$E[S_{100}] = 2E[W] - 100 = 2\left(100\frac{18}{37}\right) - 100 = -\frac{100}{37},$$

and

$$\mathrm{Var}(S_{100}) = 4\mathrm{Var}(W) = 4\left(100\frac{18}{37}\frac{19}{37}\right) = 100\frac{1368}{1369}.$$

A simulation of

$$200000$$

independent 100-spin sessions produced mean profit

$$-2.67761$$

and variance

$$99.68430.$$

The simulated mean was approximately

$$1.12$$

Monte Carlo standard errors from the theoretical value. Therefore, the simulation supports both the expected value and variance calculations.



LLM Prompts

Prompt 1: Roulette

> Consider European roulette with outcomes $0, 1, \ldots, 36$. Construct the probability space and sigma-algebra, identify common events such as red, even, and zero, and compute the expectation and variance of a one-unit bet on red.

Prompt 2: Blackjack

> Consider a simplified one-deck blackjack model in which the player and dealer hit below 17 and stand on 17 or greater. There is no splitting or doubling. Describe a rigorous probability space using ordered decks, define the payoff

> random variable, and explain why blackjack is more complicated than roulette.

Prompt 3: Bankroll Allocation

> A player has bankroll B and may bet b_R on roulette and b_B on blackjack, where $b_R, b_B \geq 0$ and $b_R + b_B \leq B$. If the expected returns per dollar are e_R and e_B, solve the problem of maximizing $b_R e_R + b_B e_B$ and discuss what happens when the expected returns are negative or positive.

Prompt 4: Five Consecutive Roulette Losses

> In European roulette there are 18 red numbers, 18 black numbers, and one zero. A player bets one unit on red for five independent spins. What is the probability that the player loses all five bets? Define the sample space and event and derive the probability.

Prompt 5: One Hundred Roulette Bets

> A player makes 100 independent one-unit bets on red in European roulette. Define the total-profit random variable and calculate its expected value and variance. Verify the expectation using a binomial model.

Discussion of the LLM

The LLM was useful for organizing the probability models and suggesting ways to verify the calculations.

For roulette, it correctly identified the finite sample space

$$\Omega_R = \{0, \ldots, 36\}$$

and the power-set sigma-algebra

$$\mathcal{F}_R = 2^{\Omega_R}.$$

It also correctly calculated

$$E[X_R] = -\frac{1}{37}$$

and suggested using Monte Carlo simulation as an independent check.

For blackjack, the LLM was useful in recognizing that an ordered deck can serve as the underlying sample point. Once the deck order and strategy are fixed, the hand becomes

deterministic.

The LLM was also useful in suggesting a recursive calculation rather than attempting to explicitly enumerate all

$$52!$$

possible ordered decks.

However, the LLM could be mathematically careless when details of the model were not specified.

For example, the original blackjack problem did not completely specify:

- whether a natural blackjack pays $3:2$;
- whether the dealer hits or stands on soft 17;
- how pushes are treated;
- how aces are valued;
- the exact player strategy.

A numerical expected value cannot be meaningfully calculated until these assumptions are fixed.

For this report I explicitly chose a simplified model in which all wins pay $+1$, pushes pay 0, losses pay -1, and both the player and dealer stand on every total of 17 or greater.

Another possible error would be treating successive blackjack card draws as independent. This would be incorrect because cards are drawn without replacement.

The recursive calculation and simulation used here update the remaining deck after every card, so this dependence is correctly represented.

The bankroll problem also illustrates why an LLM answer must be interpreted carefully. Saying that one game is better than another does not imply that it should be played.

For example, in my model,

$$e_R \approx -0.0270$$

is better than

$$e_B \approx -0.0801.$$

However,

$$0 > e_R > e_B,$$

so keeping the bankroll is better than either game.

The LLM correctly identified this once the constraint

$$b_R + b_B \leq B$$

was taken into account.

Overall, the LLM was useful as a tool for constructing models, proposing calculations, writing simulation code, and checking reasoning. However, its answers still required mathematical verification because small unstated assumptions can substantially change a probability model.

Conclusion

The roulette probability space is

$$(\Omega_R, \mathcal{F}_R, P),$$

where

$$\Omega_R = \{0, \ldots, 36\}$$

and

$$\mathcal{F}_R = 2^{\Omega_R}.$$

For a one-unit red bet,

$$E[X_R] = -\frac{1}{37} \approx -0.027027$$

and

$$\mathrm{Var}(X_R) = \frac{1368}{1369} \approx 0.999270.$$

For the simplified blackjack strategy,

$$E[X_B] \approx -0.0800918$$

and

$$\mathrm{Var}(X_B) \approx 0.8913244.$$

Both expected values are negative.

Therefore, for the actual models used in this homework,

$$E[\Pi] = -0.027027 b_R - 0.0800918 b_B.$$

The expected-value maximizing bankroll allocation is

$$b_R^* = 0, \qquad b_B^* = 0.$$

Thus the entire bankroll should be kept rather than bet.

The analytical calculations, recursive blackjack computation, grid-search optimization, and Monte Carlo simulations all support the same conclusion.

The homework also demonstrates an important lesson about using LLMs in mathematics: an LLM can be very useful for generating models and calculations, but its assumptions and conclusions should always be independently checked.